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CS-2403

B. Tech. (Third Semester)

EXAMINATION, 2024-25

**COMPUTER BASED NUMERICAL
METHOD AND STATISTICAL TECHNIQUE**

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *seven* questions. Each question carries 8 marks. $7 \times 8 = 56$

1. Given a solution of a problem as $x_a = 35.25$ with relative error in the solution at most 2%. Find to four decimal places, the range

34.5450 , 35.4550

P. T. O.

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of values within which the exact value of the solution must lie.

2. If $r = 3h(h^2 - 2)$, find the percentage error in h is 5. $\frac{5}{2}\%$

3. Use bisection method to find out the positive square root of 30 correct to four decimal places. 5.4721

4. Find the root of equation $xe^x = \cos x$ in interval $(0, 1)$ using Regula-Falsi method correct to four decimal places.

5. Construct the backward difference table from the following data :

$\sin 30^\circ = 0.5, \sin 35^\circ = 0.5736$
 $\sin 40^\circ = 0.6428, \sin 45^\circ = 0.7071$

Assuming third differences to be constant, find the value of $\sin 25^\circ$.

6. From the following table the half yearly premium for policies maturing at different

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Apply Simpson's three-eighth rule to determine the distance moved by the train in 45 seconds.

Section—C

(Long Answer Type Questions)

Note : Attempt any one question. Each question carries 20 marks. $1 \times 20 = 20$

1. Given $\frac{dy}{dx} = \frac{y-x}{y+x}$ with $y = 1$ for $x = 0$. Find y approximately for $x = 0.1$ by Euler's method.

2. For the differential equation, $\frac{dy}{dx} = -xy^2$, $y(0) = 2$, calculate $y(0.2)$ by Taylor's series method retaining four non-zero terms only.

3. Given $\frac{dy}{dx} = y - x$, $y(0) = 2$. Find $y(0.1)$ and $y(0.2)$ correct to four decimal places. Use Runge-Kutta IV order method. $2.1368, 2.2752$

P. T. O.

4. Approximate y and z by using Picard's method for particular solution of

$$\frac{dy}{dx} = x + z, \quad \frac{dz}{dx} = x - y^2, \quad \text{given that } y = 2,$$

$z = 1$ when $x = 0$.


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CS-2402

B. Tech. (Third Semester)
EXAMINATION, 2024-25
DATA STRUCTURE USING "C"

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *seven* questions in about
30 words Each question carries
8 marks. $7 \times 8 = 56$

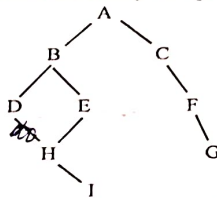
1. Write an algorithm for binary search.
2. What is circular queue ?
3. Differentiate between static and dynamic memory allocation.
4. Explain AVL tree.
5. Define B-tree.

P. T. O.

[2]

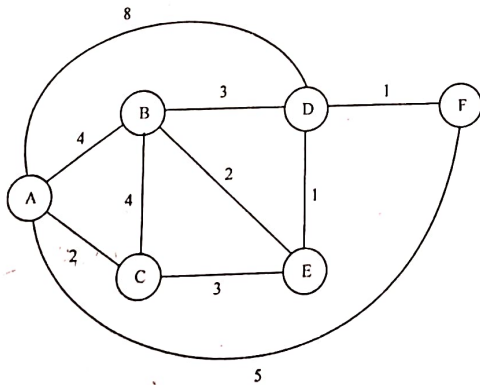
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6. Consider the binary tree given below :



Traverse the given tree using pre-order and post-order traversal.

7. Write a pseudo code for deleting an element from a binary search tree.
8. Draw the minimum cost spanning tree for the graph given below :



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9. What do you understand by the term algorithm ?
10. Write an algorithm for bubble sort.

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions in about **150** words. Each question carries 12 marks. $2 \times 12 = 24$

1. Explain stack. Also explain various applications of stack.

2.	Symbol	Frequency
	A	10
	B	50
	C	90
	D	30
	E	40
	F	3

Handwritten note:
 Parathesise
 inside - possible
 binary search tree
 Brackets
 BFS, DFS

Construct Huffman tree for the above message, assign code to each symbol and find the compression ratio if ASCII 7.0 is used to store this text.

P. T. O.

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3. Define link list and also explain its various types with suitable example.
4. Sort this array with the help of quick sort and also find its average, worst and best case complexity :

12, 6, 9, 25, 10, 18, 30, 7

Section—C

(Long Answer Type Questions)

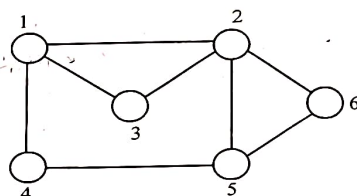
Note : Attempt any *one* question in about **300** words. Each question carries 20 marks. $1 \times 20 = 20$

1. Define binary tree and also explain almost complete binary tree and complete binary tree with suitable example.
2. What is sequential search ? Give its algorithm.
3. Define AVL tree. Explain the following rotations with example :
 - (i) RL rotation
 - (ii) LL rotation
 - (iii) RR rotation
 - (iv) LR rotation

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4. Apply BFS and DFS on the following graph :



and also give the meaning of path matrix.

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Handwritten notes and diagrams in the bottom right corner, including a small tree diagram and some scribbles.

Ritika Singh

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CS-2404

B. Tech. (Third Semester)

EXAMINATION, 2024-25

COMPUTER ORGANIZATION

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *seven* questions in about
30 words. Each question carries
8 marks. $7 \times 8 = 56$

1/ Define Cache memory in brief.

P. T. O.

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2. Define synchronous mode data transfer.
3. What is Tristate buffer ?
4. Define Wide-Branch Addressing.
5. State General Purpose Register with diagram.
6. Differentiate between vector and non-vector interrupt.
7. What is BUS ? Explain with suitable example.
8. What is Input-Output Interference ?
9. Differentiate between SRAM and DRAM.
10. What is Multiplexer and Demultiplexer ?

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions in about 150 words. Each question carries 12 marks. $2 \times 12 = 24$

1. Discuss the use of microoperation in computer organization with their types.

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2. What is addressing mode in organization ? Also explain any *three* types of addressing mode.
3. Explain Booth's algorithm and prefer multiplication (-5×-4) using Booth's algorithm.
4. Why mapping is used in a computer organization ? Also explain all types of mapping.

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question in about 300 words. Each question carries 20 marks. $1 \times 20 = 20$

1. Why is arbitration method used in computer organization ? Explain difference between Daisy chain and Parallel method.

P. T. O.

2. (a) Explain virtual memory.
(b) Explain the difference between 2D RAM and 2½ RAM organisation.
3. Explain Pipelining in brief.
4. (a) Explain Hazard in brief.
(b) Explain Flynn classification.

P. T. O.

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CS-2401

B. Tech. (Third Semester)

EXAMINATION, 2024-25

DISCRETE STRUCTURE

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *seven* questions. Each question carries 8 marks. $7 \times 8 = 56$

1. Two finite sets have m and n elements. The total number of subsets of the first set is 56 more than the total number of subsets of the second set. Find the values of m and n .

P. T. O.

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2. If $A = \{x : x \in \mathbb{N}, x \text{ is a factor of } 6\}$ and
 $B = \{x : x \in \mathbb{N}, x \text{ is a factor of } 8\}$
 then find (i) $A \cup B$ (ii) $A - B$

3. Prove that :

$$(i) (A \cup B)' = A' \cap B'$$

$$(ii) (A \cap B)' = A' \cup B'$$

4. In a city of 1000 families, it was found that 40% families buy newspaper A, 20% families buy newspaper B, and 10% buy C. Only 5% families buy A and B, 3% buy B and C and 4% buy A and C and 2% families buy all the three newspapers. Find the number of families which buy :

(i) A only

(ii) B only

5. If a set have n elements, then prove that total number of Reflexive Relation Possible are (different-different).

$$2^{n(n-1)}$$

[3]

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6. Let A be the set $\{1, 2, 3\}$, define the following types of binary relation on A :

- (i) A relation that is both symmetric and anti-symmetric.
 (ii) A relation that is neither symmetric nor anti-symmetric.

7. Let :

$$X = \{1, 2, 3, \dots, 7\}$$

$$\text{and } R = \{x, y\} : (x - y) \text{ is divisible by } 3\}.$$

Show that R is an equivalence relation.

8. If :

$$g(x) = 1 - x$$

$$\text{and } h(x) = \frac{x}{x-1},$$

then find the value of $g(h(x)) / h(g(x))$.

P. T. O.

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9. Prove that :

$$2^n < n!$$

for every positive integer n with $n \geq 4$,
using mathematical induction.

10. Consider the operation $*$ defined on $\mathbb{Z}$, the
set of integers, as :

$$a * b = a + b - 3ab$$

Determine whether $*$ is commutative and
associative.

Section—B

(Short Answer Type Questions)

Note : Attempt any two questions. Each
question carries 12 marks. $2 \times 12 = 24$

- Find the greatest lower bound and the least
upper bound of the set $\{2, 3, 6\}$ if they
exist, in the poset $(D_{24}, |)$. Also draw the
Hasse diagram of $(D_{24}, |)$.
- Let $(\{a, b\}, *)$ be semi group where
 $a * a = b$ show that :
(i) $a * b = b * a$
(ii) $b * b = b$

Clear
Answer

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3. For any two elements a and b of Boolean
algebra B prove that :

$$(i) (a + b)' = a' \cdot b', \forall a, b \in B$$

$$(ii) (a \cdot b)' = a' + b', \forall a, b \in B$$

4. (a) Change the function f ,
$$f = uv + u'v + u'v'$$

from DN form to CN form where DN
stands for Disjunctive form and CN
stands for conjunctive Normal form.

(b) Simplify the Boolean function by using
laws of Boolean algebra :

$$xy + x'y + x'y'$$

Section—C

(Long Answer Type Questions)

Note : Attempt any one question. Each
question carries equal marks. $1 \times 20 = 20$

- Define the following :
(a) Monoid

P. T. O.

[6]

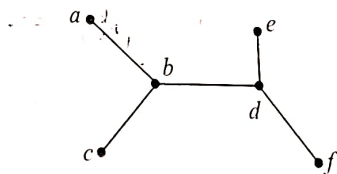
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(b) Ring

(c) Totally ordered set

(d) Prim's algorithm for minimum spanning tree

2. (a) Find the eccentricity of all vertices, radius and centre of the tree in given figure :



(b) Represent the expression :

$$(((x+2) \uparrow 3) * (y - (3+x))) - 5$$

using a binary tree and write the expression in postfix notation and prefix notation.

3. Define the following with example :

(i) Lattice

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(ii) Bounded Lattice

(iii) Distributive Lattice

(iv) Complemented Lattice

4. (a) Consider the following conditional statement :

P : If the floods destroy my house or the fires destroy my house, then my insurance company will pay me. Write the converse, inverse and contrapositive of the statement.

(b) Show that :

$$\{[(p \vee q) \rightarrow r] \wedge (\neg p)\} \rightarrow (q \rightarrow r)$$

is a Tautology without using Truth table.

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